HDOC Day-49

C Programming Activity

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Aim

Write a C program to find the pivot integer x between 1 and n such that the sum from 1 to x is equal to the sum from x to n, using pointers to pass n to the function.

C Program

```
#include <stdio.h>

void pivot_integer(int *n)
{
    int x, i;
    int left_sum, right_sum;

    for (x = 1; x <= *n; x++)
    {
        left_sum = 0;
        right_sum = 0;

        for (i = 1; i <= x; i++)
            left_sum = left_sum + i;

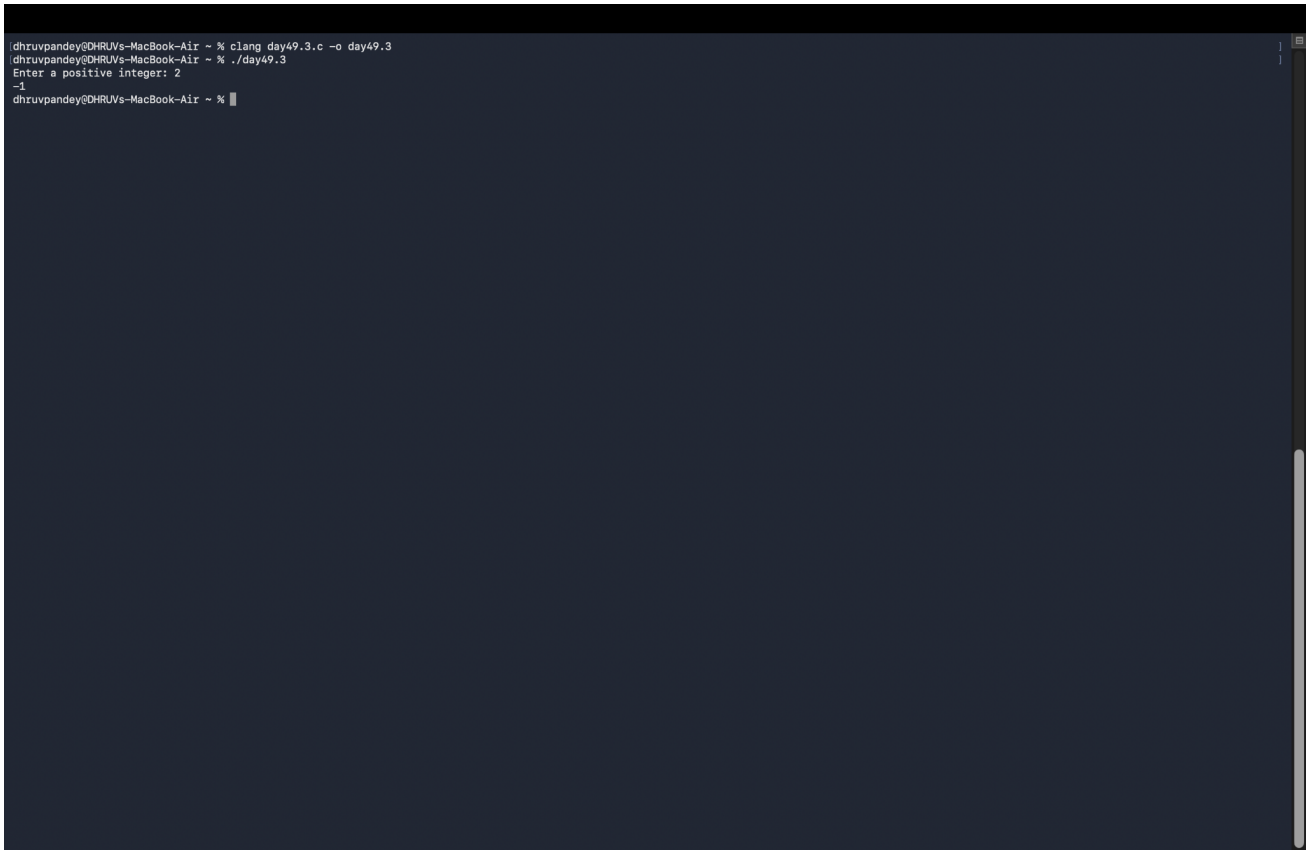
        for (i = x; i <= *n; i++)
            right_sum = right_sum + i;

        if (left_sum == right_sum)
        {
            printf("Pivot integer: %d\n", x);
            return;
        }
    }
    printf("-1\n");
}

int main()
{
    int n;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    pivot_integer(&n);
    return 0;
}
```

Execution

Program compiled and executed in the terminal. For **n = 2**, the program correctly displayed **-1** because no pivot integer exists.



Result: The program successfully finds the pivot integer using pointer-based passing and displays -1 when no pivot integer exists.